



Terms of Reference – CAN 2025 Fanzine Application



Application Analysis

Overview

Gamification application for customer engagement during the Africa Cup of Nations (AFCON) 2025, with a dual participation mechanism through unique QR codes distributed in partner cafés.

Key Critical Points

1. Scalability & Performance

- Massive traffic spikes during matches.
- Simultaneous management of thousands of QR scans.
- Need for a CDN for static assets.
- Database optimized for intensive read/write operations.

2. Security

- One-time-use QR codes (anti-fraud).
- Traceability of scans by café/server.
- Protection against spam in predictions.
- Mandatory server-side validation.

3. User Experience

- Mobile-first interface (90% of scans from smartphones).
 - Partial offline functionality.
 - Immediate feedback after scanning.
 - Notifications for raffle/prediction results.
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Detailed Terms of Reference

Recommended Technical Architecture

Proposed Production-Ready Stack

Frontend

- Next.js 14 (App Router) – SSR/SSG for performance.
- TypeScript – Type safety.

- Tailwind CSS – Fast UI development.
- React Query – Smart caching.
- Vercel – Deployment with edge functions.

Backend

- Node.js + Express / NestJS.
- PostgreSQL (main database) – Complex relationships.
- Redis – Cache + sessions + rate limiting.
- Prisma ORM – Type-safe queries.

Infrastructure

- AWS/GCP or Vercel + Supabase (simplicity).
- Cloudflare CDN for assets.
- Queue system (Bull/BullMQ) for asynchronous processing.
- Monitoring: Sentry + Datadog.

Cost-Effective Alternative

- Supabase (Backend-as-a-Service) – Auth + DB + Realtime.
- Vercel (Frontend + API Routes).
- Drastically reduces costs and complexity.

Detailed Features

A. User Journey – First Visit

- 1. Scan QR Code**
 - o Validation of code (unique, not expired).
 - o Redirect to registration form.
- 2. Registration/Login**
 - o Phone number (+ SMS OTP).
 - o Full name.
 - o Acceptance of Terms & Conditions.
- 3. Choose Participation Type**
 - o Modal with two visual options.
 - o Rules explained.
 - o Final choice (cannot be changed).
- 4. Personalized Dashboard**
 - o Based on selected type.

B. Football Supporter – Features

After each scan:

-  Automatic raffle ticket generated.
-  Unique number displayed.
-  Total accumulated tickets.

-  Date of draw.

Supporter dashboard:

- My raffle tickets (list).
- Scan history (café, date).
- Draw status.
- Notifications if winner.

Raffle rules:

- 1 scan = 1 ticket.
- Weekly or phase-based draws.
- Prizes defined (to be specified).

C. Football Analyst – Features

After each scan:

-  Bonus point added (e.g., +5 points).
-  Position in ranking.
-  Upcoming matches (open for predictions).

Analyst dashboard:

- Match calendar.
 - Filters: Phase, group, date.
 - Status: To predict / Completed.
 - Prediction deadline (2h before kick-off).
- Make a prediction.
 - Exact score (e.g., 2–1).
 - Validation before deadline.
 - Modifiable until deadline.
- Global ranking.
 - Top 100 visible.
 - Filters: Global, by region, friends.
 - My position.
- My statistics.
 - Total points.
 - Correct predictions.
 - Best streak.
 - Progress charts.

Scoring system:

- Exact score: 5 points.
- Correct result (win/draw): 3 points.
- Correct goal difference: +1 point.
- Bonus points per scan: +2–5 points.

D. Admin Dashboard

Overview:

- Total participants (real-time chart).
- Supporter/Analyst distribution.
- Scans per day/hour.
- Top active cafés.

QR Code management:

- Bulk generation (by café).
- Export printable PDF.
- Deactivation of lost codes.
- Usage statistics.

Raffle management:

- Entry list.
- Launch random draw.
- Mark winners.
- Export winners list.

Match management:

- Import AFCON calendar.
- Real-time score updates.
- Automatic points calculation.
- Prediction lock.

Analytics:

- Return rate (rescans).
 - Best-performing cafés.
 - Engagement by region.
 - Data export (GDPR compliant).
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